

At-Risk Patient Logic

Day 6 — Vital Sign: Pulse (Heart Rate)

Overview

This pseudocode defines a function that evaluates a patient's pulse (heart rate in beats per minute) and flags them as at-risk if their pulse falls outside the defined safe range. Patients with a pulse below 20 bpm or above 250 bpm are considered critical.

Pseudocode

```
FUNCTION flag_at_risk_by_pulse(patient_id, pulse_bpm):  
  
  SET at_risk = FALSE  
  SET reason = "Normal"  
  
  IF pulse_bpm < 20 THEN  
    SET at_risk = TRUE  
    SET reason = "Critical Low: pulse below 20 bpm"  
  
  ELSE IF pulse_bpm > 250 THEN  
    SET at_risk = TRUE  
    SET reason = "Critical High: pulse above 250 bpm"  
  
  END IF  
  
  RETURN patient_id, pulse_bpm, at_risk, reason  
  
END FUNCTION
```

Threshold Definitions

Condition	Pulse Range	Rationale
Critical Low	< 20 bpm	Near-absent cardiac output; immediate intervention required

Normal	20 – 250 bpm	Functional range covering extreme athletic/pediatric cases
Critical High	> 250 bpm	Exceeds survivable arrhythmia rates (e.g. ventricular flutter)

Clinical Note

Thresholds are set at extreme boundaries (20–250 bpm) to capture only the most critical deviations, allowing the system to trigger high-priority alerts without excessive false positives. Standard bradycardia (<60 bpm) and tachycardia (>100 bpm) may be incorporated in a secondary warning tier.